

Case Study: Odisha Heat Action Plan

Institutional Disaster Management, Humidity-Indexed Alerts, and Multi-Sectoral Adaptation

Policy Analysis Report

July 30, 2026

Abstract

This case study analyzes the evolution, strategic mechanics, quantitative outcomes, and challenges of the Odisha Heat Action Plan. Triggered by a catastrophic heatwave in 1998 that claimed over 2,000 lives, Odisha pioneered sub-national disaster management infrastructure in India by establishing the Odisha State Disaster Management Authority (OSDMA). By integrating humidity-adjusted early warnings, decentralized hydration networks (*Jal Seva Kendras*), legally enforced mandatory labor stoppages, and ward-level spatial hotspot mapping, Odisha has achieved a dramatic, multi-decade reduction in extreme heat mortality.

1 Background & Catalyst Event

1.1 The 1998 Heatwave Crisis

In May 1998, the eastern coastal state of Odisha experienced an unprecedented summer heatwave that caused **2,042 reported fatalities**. The crisis was exacerbated by the interplay between elevated ambient temperatures and high relative humidity along the Bay of Bengal, coupled with a lack of institutional early warning systems and low public awareness regarding heatstroke risks.

1.2 Institutional Paradigm Shift

The 1998 tragedy led to the establishment of the **Odisha State Disaster Management Authority (OSDMA)** in 1999—making Odisha the first state in India to establish a dedicated autonomous disaster management agency. Over two decades, OSDMA transitioned from reactive emergency response to a proactive, multi-departmental **State Heat Action Plan** in collaboration with:

- **IMD:** India Meteorological Department
- **BMC / Municipalities:** Local Urban Civic Bodies
- **Health & Family Welfare Department**
- **Labour & Employees' State Insurance Department**

2 Strategic Architecture & Core Pillars

Odisha's Heat Action Plan operates through inter-agency coordination spanning five structural pillars:

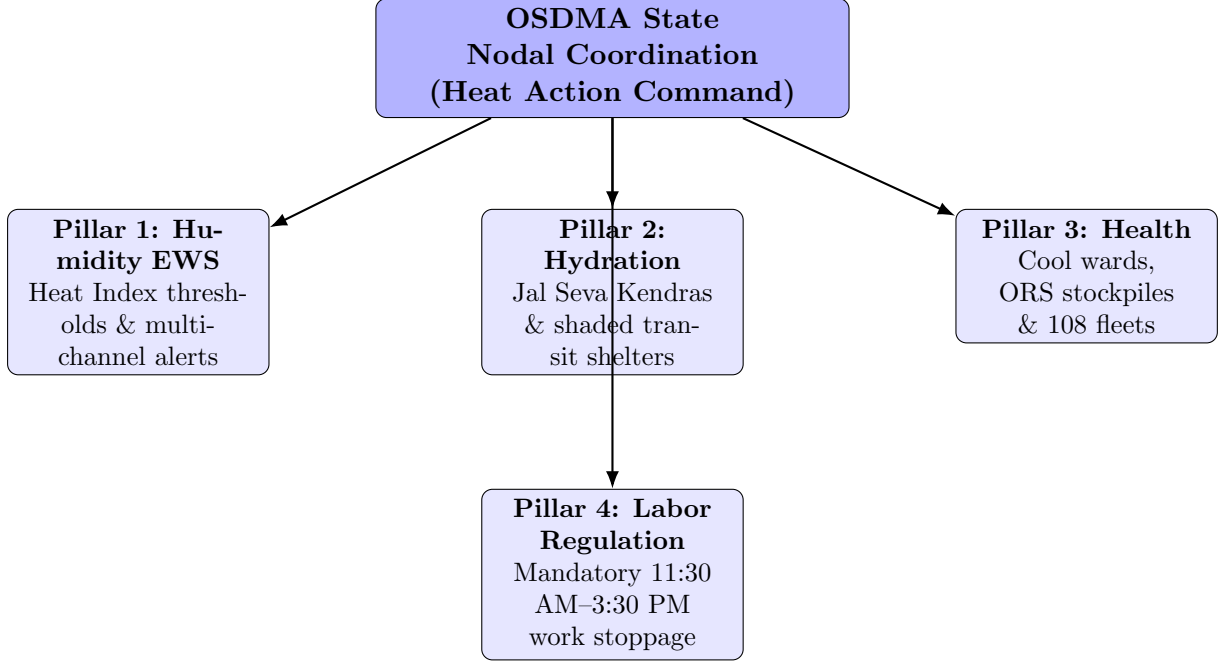


Figure 1: Institutional Architecture of the Odisha Heat Action Plan.

2.1 Pillar 1: Humidity-Indexed Early Warning System (EWS)

Unlike inland dry regions, Odisha’s coastal geography subjects metropolitan zones like Bhubaneswar and Cuttack to high relative humidity ($RH \approx 70\% - 85\%$). The plan uses a **Heat Index (HI)** metric combining dry-bulb temperature (T) and humidity (RH) to assess apparent temperature:

$$\text{Heat Alert Risk} = f(T_{\text{air}}, RH) \quad (1)$$

When the calculated Heat Index crosses danger thresholds, OSDMA issues color-coded **Yellow**, **Orange**, and **Red** warnings via broadcast media, direct SMS pushes, digital LED transit boards, and automated messaging.

2.2 Pillar 2: Public Hydration Infrastructure (*Jal Seva Kendras*)

- **Decentralized Kiosks:** Thousands of state-funded temporary drinking water kiosks (*Jal Seva Kendras*) are set up at transit hubs, market squares, highway intersections, and rural bus stops, managed by Women’s Self-Help Groups (SHGs) and civic bodies.
- **Shade Canopy Structures:** Installation of green shade tarps over traffic intersections to protect commuters and motorcyclists from direct solar exposure during peak afternoon hours.

2.3 Pillar 3: Mandatory Labor Restrictions and Schedule Shifts

- **Work Stoppage Enforcement:** Enforcement of legal restrictions on outdoor manual labor (including construction sites, brick kilns, stone quarries, and MGNREGA rural work schemes) between **11:30 AM and 3:30 PM** during heatwave warnings.
- **School Schedule Adjustments:** Educational institutions and municipal administrative services transition to early morning operational hours (06 : 30 AM – 11 : 30 AM) during peak summer months.

2.4 Pillar 4: Health System Readiness

- **Dedicated Cool Wards:** Creation of climate-controlled emergency heat Wards in Urban Primary Health Centers (UPHCs), District Headquarter Hospitals (DHHs), and state medical colleges.
- **Paramedic Protocols:** Stockpiling of Oral Rehydration Salts (ORS), IV saline fluids, and ice packs across the state’s 108 ambulance network and community health worker kits.

3 Key Innovations: Hotspot Mapping & Wildlife Adaptation

1. **Ward-Level Micro-Hotspot Mapping:** In cities like Bhubaneswar, OSDMA leverages remote sensing Land Surface Temperature (LST) data paired with socio-economic vulnerability metrics (slum density, outdoor laborer demographics) to allocate shade and water infrastructure to high-vulnerability urban wards.
2. **Ecosystem & Wildlife Protection:** Odisha extends heat plan mitigation into protected natural reserves (such as Similipal Tiger Reserve and Chandaka Sanctuary) by creating artificial water bodies, check dams, and tanker fill-stations to prevent wild animal dehydration and combat forest fire risks during extreme dry spells.

4 Seasonal Implementation Framework

Odisha structures its operational workflows along three distinct seasonal phases:

Table 1: Seasonal Phase Execution Cycle of the Odisha Heat Action Plan

Phase	Timeline	Primary Action Items
Phase I: Pre-Summer	Jan – Mar	Inter-departmental planning meetings; auditing healthcare ORS/IV inventories; deploying <i>Jal Seva Kendras</i> ; briefing labor inspectors.
Phase II: Peak Heat	Apr – Jun	Tracking daily IMD Heat Index alerts; enforcing the 11:30 AM–3:30 PM work ban; operating cooling wards; tracking heatstroke morbidity data.
Phase III: Post-Heat	Jul – Sep	Conducting mortality data audits; assessing water kiosk coverage; refining district and ward heat vulnerability maps.

5 Public Health Impact and Critical Challenges

5.1 Quantifiable Success

- **Dramatically Reduced Mortality:** From **2,042 reported fatalities in 1998**, annual direct heatstroke fatalities dropped significantly into low double digits or near-zero in structured plan years, despite rising baseline temperatures.
- **Institutional Integration:** Heat management budgets are embedded directly across line departments (Labour, Health, Urban Development) rather than relying solely on post-disaster funding mechanisms.

5.2 Operational Challenges

- **Wet-Bulb Temperature Risks:** Coastal humidity elevates wet-bulb temperatures ($T_{\text{wet-bulb}} \geq 32^{\circ}\text{C}$), which impairs the human body’s evaporative cooling efficiency through sweating even when dry-bulb air temperatures remain moderate.
- **Informal Sector Compliance:** While organized construction firms abide by mandatory work bans, enforcing shift adjustments remains difficult for street vendors, gig workers, and informal daily-wage earners who face immediate financial loss from taking work breaks.

6 Conclusion

The Odisha Heat Action Plan demonstrates how institutional disaster planning, humidity-adjusted early warnings, and coordinated public health measures can reduce mortality caused by extreme heat events. As climate change increases both ambient temperatures and coastal humidity, Odisha’s model offers valuable policy insights for sub-national disaster management across heat-vulnerable regions worldwide.